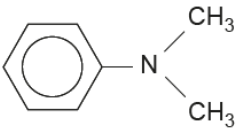


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		award (1) for either of following - sodium nitrite / nitrate(III) / NaNO_2 and HCl / hydrochloric acid - nitrous acid / nitric(III) acid / HNO_2 / HONO	1			1		1
		(ii)				1		1		
		(iii)		$f = \frac{c}{\lambda}$ (1) $7.32 \times 10^{14} \text{ Hz}$ (1)	1	1		2	2	
	(b)	(i)		number of moles = $\frac{0.0075 \times 250}{1000} = 1.875 \times 10^{-3}$ (1) mass required = $272 \times 1.875 \times 10^{-3} = 0.51 \text{ g}$ (1)		2		2	1	
		(ii)		$\frac{1.44}{1.03} = \frac{k \times 0.0096}{k \times c}$ (1) $c = \frac{0.0096 \times 1.03}{1.44} = 6.87 \times 10^{-3} / 0.00687 / 0.0069 \text{ mol dm}^{-3}$ (1) accept alternative method $k = 150$ (1) $c = 6.87 \times 10^{-3} / 0.00687 / 0.0069 \text{ mol dm}^{-3}$ (1)		2		2	2	